

$$(15.19) \quad \sigma_y^2 = \left(\frac{1}{4}\right) (\sigma_{x_1}^2 + \sigma_{x_2}^2)$$

if the case that the distributions were identically distributed with expected value and variance of μ_x and σ_x^2 , each partner would face the same expected value as before, μ_x . But, the variance of their individual earnings would be $(\sigma_x^2 + \sigma_x^2)/4 = \sigma_x^2/2$, half of what it was before without combining their businesses. Furthermore, the standard deviation of the earnings each partner would face would be:

$$(15.20) \quad \sqrt{\frac{\sigma_x^2}{2}} = \frac{\sigma_x}{\sqrt{2}}$$

And if n partners joined together, then they would each face the same expected value as before, but the variance each partner would receive is σ_x^2/n . We now illustrate these important results.

Assume that business one's earnings are determined by outcomes associated with the toss of a fair coin. If the outcome of the coin toss is tails, the firm pays (loses) \$5,000. If the toss is a heads, the firm wins \$8,000. Thus, the firm wins either \$8,000 or loses \$5,000 and earns on average $(.5)(-5,000) + (.5)(8,000) = \$1,500$.

The standard deviation of this risky outcomes is:

$$(15.21) \quad \sqrt{(.5)(-5,000 - 1,500)^2 + (.5)(8,000 - 1,500)^2} = \$6,500$$

Furthermore, assuming a normal distribution, 68% of the time, the average outcome will be between the mean and plus or minus one standard deviation: $(1,500 + 6,500) = \$8,000$ and $(1,500 - 6,500) = -\$5,000$.

Now suppose that two persons decide to combine their operations and share the average of the outcomes. Then the possible outcomes of two coin tosses are two heads (H, H) which earns on average $\$16,000 / 2 = \$8,000$ and occurs with a probability of .25; two tails (T, T) which earns on average $-\$10,000 / 2 = -\$5,000$ and occurs with a probability of .25, and one head and one tail (H, T) or one tail and one head (T, H) which both earn on average $\$3,000 / 2 = \$1,500$ and each occurs with a probability of .25. The expected value for each of the two players can now be expressed as:

$$(15.22) \quad (.25)(\$8,000) + (.25)(-\$5,000) + (.25)(\$1,500) + (.25)(\$1,500) = \$1,500$$

The two players now receive on average the same as before, \$1,500, but consider the standard deviation of the average outcome: